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# SDG 12.3 Food Waste Analytics Dashboard – Ireland

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# 1. Introduction

Bright Insight Analytics Services (BIAS) has contracted our team to assist it in its new dashboarding division, which deals with tracking the progress towards the UN Sustainable Development Goals (SDGs). This department is run on the project-based model and provides data-driven visual information to public, private, and nonprofits on the organisational, national, and global levels. In this engagement, the task is to create an interactive dashboard based on trusted sources of open data to track the progress in relation to the chosen SDG indicators to make evidence-based assessment and make an appropriate decision with consideration of the priorities of sustainable development.

# 2. Client Organisation and Context

The client that we are dealing with is GreenPlate Ireland, which is a national nonprofit association of major retailers, hospitality groups, and agricultural producers. GreenPlate is seeking the services of BIAS to develop a data-driven dashboard to track and report on the progress made by Ireland in minimizing food waste throughout the supply chain. They need an open and transparent instrument that keeps the stakeholders on their toes, educates their citizens, and convinces them to invest more in waste-cutting technologies. The dashboard will be the source of evidence that is necessary to focus on the interventions by means of evidence and prove society and policymakers of the greater benefits of their work.

# 3. SDG Focus and Targets

The mission of GreenPlate is closely related to SDG 12 (Responsible Consumption and Production), Target 12.3 in particular: "By 2030, 50 per cent per capita global food waste in the retail sector and the consumer sector and half food waste across the production and supply chains, including post-harvest food waste (United Nations, 2025) . This dashboard will also shine light into progress on two goals that are of great essential interconnection:

- *SDG 2 (Zero Hunger):* Minimizing food waste will open resources and food to achieve Target 2.1, which is to provide access to safe and nutritious food to everyone (United Nations, 2025).
- *SDG 13 (Climate Action):* Because the concept of decomposing food waste is one of the primary contributors to methane, it will directly help Target 13.2 to include climate change in national policies and mitigate emissions (United Nations, 2025) .

Through monitoring SDG 12.3, the dashboard will give an overall picture of how Ireland has done at developing a more efficient and equitable, as well as low-carbon food system.

# 4. Indicator Selection and Rationale

The primary indicator for SDG 12.3 is "Food Waste per Capita (kg)," disaggregated by sector (e.g., household, retail, food service) (United Nations, 2025). This metric is the most direct and adequate measure of progress against the target's core aim of halving per-capita waste.

To contextualize this data and demonstrate its broader impact, we will integrate two key supplementary measures:

- Carbon Dioxide Equivalent (CO 2 e) in Food Waste: This measures the ton of food waste in emissions which has a direct correlation to SDG 13.2 and displays its environmental price (United Nations, 2025).
- Potential Population Fed on Food Wasted: It is a computed value to justify SDG 2.1. We can re-run the figures powerfully by transforming the overall tonnage of edible food waste into an estimated number of adult daily dietary energy requirements (e.g., 2,500 kcal) (United Nations, 2025). The dashboard will not display the tonnes wasted, but rather display This wasted food could have provided X people with a year of food. It is relatable and emphasizes that the food system is incredibly socially inefficient, which makes it a powerful change story.

# 5. Qualitative Description of Datasets

The dashboard analysis will be based on various connected sets of data, most of which will be obtained through the official SDG Data Hub (EPA, 2023), which is a compilation of data provided by various national institutions in Ireland such as the Central Statistics Office (CSO) and the Environmental Protection Agency (EPA). Alternative global data is obtained through the United Nations agencies to give a background on emissions and food security.

## 5.1 Core Food Waste Datasets

The Food Waste data is the most essential in the process of following SDG 12.3. It gives the scope of annual numbers 2018-2023, according to sector: primary production, processing, retail/distribution, restaurants, and households (EPA, 2023). It was developed by the EPA, and it is used to track the Irish progress with the national and EU targets of food waste reduction. One of the main weaknesses is that the primary production sector data is not available in 2018 and 2019, and it is important because the primary production sector can be a significant contributor to the total waste footprint. The dataset of Compost & Anaerobic Digestion is

utilized to comprehend the treatment of organic waste. It is covering 2005-2022 and follows the tonnages of biowaste that are covered by these recovery processes. Nevertheless, key metrics lack data, as 2005-2016 have a blank in the analysis of organic recovery trends in the first part of the timeframe.

To place food waste within Ireland's broader waste landscape, three datasets are used:

- **National Total Waste Generation:** This EPA dataset (2012–2023) maps Ireland's total material waste footprint by sector and type. While it contains many categories, over 80% of the fields for specific waste types are empty in most years, preventing detailed compositional analysis and forcing a reliance on high-level total generation figures (EPA, 2023).
- **Municipal Waste Treatment**: This EPA dataset (2001–2023) tracks how municipal waste is managed (recycled, landfilled, incinerated for energy). It suffers from temporal fragmentation; for example, the "municipal waste generated" field is entirely missing from *2001–2009*, and other key metrics have gaps in 2013, 2015, and 2023 (EPA, 2023).
- **Household Waste**: This dataset (2010–2022) estimates waste generated by homes. A major structural gap is the missing "estimated total household waste" for 2010–2017, forcing inferences to be made from "managed waste" figures only. Data on waste streams like recycling and organics only begins in 2015, preventing a full timeline analysis of household recycling behaviour (EPA, 2023).

## 5.2 Supporting National Waste Context

To quantify the cross-cutting impact of food waste on climate and hunger, two international datasets are integrated:

- **UNFCCC Emissions**: This dataset contains greenhouse gas emissions by country and pollutant. It requires filtering for Ireland and converting units from Gigagrams (Gg) to tonnes. It will be used to model the emissions savings (SDG 13.2) achieved by reducing food waste (EEA, 2025).
- **FAO Food Production & Population**: This dataset provides food production indices and population figures. A preliminary challenge is that it lacks a direct "total production" field, requiring aggregation and scaling. The population data, however, is complete and consistent, and will be used alongside food waste data to calculate the per capita waste metric for SDG 12.3 and to model the potential population that could be fed from wasted food (SDG 2.1) (FAO, 2023; FAO, 2023).

## 5.3 Preliminary Issue Assessment

| Issuer Type | Dataset Name | Impact | Plan to Handle |
|---|---|---|---|
| Missing Data (Fields) | National Waste | Cannot perform granular waste | Interpolation |

| | | | |
|---|---|---|---|
| | | composition modelling. | |
| Missing Year Data | Municipal Waste Treatment; Compost & AD | Creates gaps in time-series analysis and trend modelling. | Adding the missing years and interpolating corresponding fields. |
| Unit Inconsistency | UNFCCC Emissions | Data not directly comparable with other tonnes-based datasets. | Unit conversion from Kilo Tonnes to Tonnes |

# 6. Quantitative Analysis for the Dataset using descriptive statistics

The quantitative analysis focuses on measures of central tendency (mean, median), dispersion (standard deviation, range, interquartile range), and data quality (missingness, outliers) for the key variables in each dataset.

## 6.1 EPA Municipal Waste Dataset

This data indicates the municipal waste indicators in Ireland in the year 2001 to 2023. The overall wastes produced are around 2.9 million tonnes with an average per-capita waste of 605 kg. Most variables have a similar median value as the mean, which implies approximately symmetric distributions, but some of the recycling measures have a small positive skew (mean > median). Standard deviations denote a moderate variability in the variables of total waste (approximately 650k tonnes) and landfill (approximately 200k tonnes), whereas organic waste and recycling have higher relative variability (approximately 140k-250k tonnes) (EPA, 2023). The margins of per-capita generation are slim (560-640 kg), however, the margins of organic waste are diverse (22k-487k tonnes).

| | Year | Generated_tonnes | Landfill_tonnes | Energy_Recovery_tonnes | Organic_tonnes | Recycled_excl_compost_tonnes | Total_Recycled_tonnes | Generated_per_capita_kg |
|---|---|---|---|---|---|---|---|---|
| count | 23.00000 | 1.100000e+01 | 2.000000e+01 | 1.300000e+01 | 20.000000 | 2.000000e+01 | 2.000000e+01 | 11.000000 |
| mean | 2012.00000 | 2.916506e+06 | 1.256743e+06 | 7.847573e+05 | 170925.137436 | 8.379052e+05 | 1.008830e+06 | 604.545455 |
| std | 6.78233 | 2.127129e+05 | 6.534203e+05 | 5.348771e+05 | 141606.781439 | 1.833606e+05 | 2.541381e+05 | 24.233711 |
| min | 2001.00000 | 2.619023e+06 | 4.180291e+05 | 8.857400e+04 | 22000.000000 | 2.835540e+05 | 3.055540e+05 | 560.000000 |
| 25% | 2006.50000 | 2.765604e+06 | 5.317228e+05 | 1.960000e+05 | 53907.500000 | 8.277863e+05 | 9.695348e+05 | 585.000000 |
| 50% | 2012.00000 | 2.846115e+06 | 1.419786e+06 | 8.808490e+05 | 131606.000000 | 8.620860e+05 | 1.063360e+06 | 620.000000 |
| 75% | 2017.50000 | 3.128020e+06 | 1.849934e+06 | 1.312957e+06 | 245696.750000 | 9.163125e+05 | 1.132120e+06 | 620.000000 |
| max | 2023.00000 | 3.210220e+06 | 2.014797e+06 | 1.412589e+06 | 487594.000000 | 1.081147e+06 | 1.312563e+06 | 640.000000 |

*Figure 1: Summary statistics for EPA Municipal Waste Dataset*

A significant challenge is missing data: *Generated_tonnes* and *Generated_per_capita_kg* are missing for ~61% of the years. Outliers are minimal, detected only in recycling metrics (*Recycled_excl_compost_tonnes* had 5 outliers).

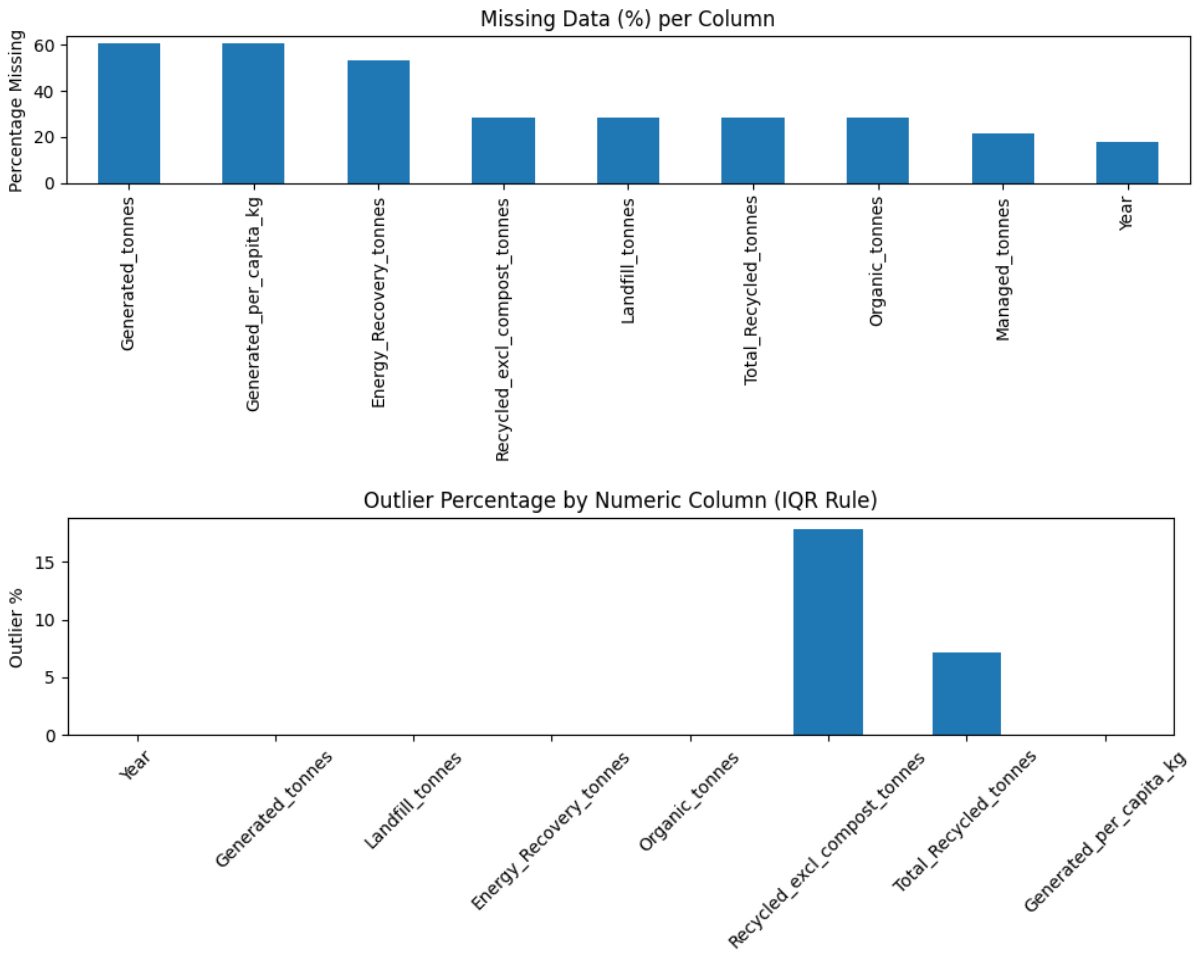



*Figure 2: Missing % and Outlier % for EPA Municipal Waste Dataset*

## 6.2 EPA Household Dataset

This data, starting in 2010 and continuing into 2022, has a total household waste of a mean of about 1.55 million tonnes (median of about 1.45 million), implying that there are some years with larger amounts of waste, drawing the mean out. Per-capita production is less erratic, the average is equal to about 319 kg, and the median equals about 311 kg. The overall variance of waste is moderate (Standard deviations of 184k tonnes) (EPA, 2023).

| | Year | Generated_tonnes | Unmanaged_tonnes | Managed_tonnes | PerCapita_kg | BlackBin_tonnes | Recyclables_tonnes | Organic_tonnes |
|---|---|---|---|---|---|---|---|---|
| **count** | 13.00000 | 1.300000e+01 | 13.000000 | 1.300000e+01 | 13.000000 | 8.00000 | 8.000000 | 8.000000 |
| **mean** | 2016.00000 | 1.546019e+06 | 14641.307681 | 1.531378e+06 | 318.543685 | 685995.16600 | 255257.636250 | 155753.060500 |
| **std** | 3.89444 | 1.836744e+05 | 20106.230713 | 1.693440e+05 | 22.325880 | 28775.97361 | 7651.492475 | 41969.162226 |
| **min** | 2010.00000 | 1.362900e+06 | 0.000000 | 1.362900e+06 | 296.688944 | 634824.62000 | 243547.830000 | 102003.254000 |
| **25%** | 2013.00000 | 1.416444e+06 | 0.000000 | 1.416444e+06 | 306.086379 | 672098.07000 | 252303.062500 | 124926.857500 |
| **50%** | 2016.00000 | 1.450727e+06 | 0.000000 | 1.450727e+06 | 310.796317 | 690493.79500 | 253955.865000 | 148208.300000 |
| **75%** | 2019.00000 | 1.622008e+06 | 31699.999850 | 1.573348e+06 | 317.303217 | 703548.93850 | 258372.395000 | 198094.180000 |
| **max** | 2022.00000 | 1.885397e+06 | 48660.000000 | 1.853697e+06 | 368.535467 | 722910.50000 | 267590.420000 | 208718.840000 |

*Figure 3: Summary Statistics for EPA Household Dataset*

There is a significant problem of missingness, with the Generated tonnes and Unmanaged tonnes being missing in about 62 percent of the record. PerCapita, kg had outliers (3 outliers) and Multiplex, tonnes (2 outliers).

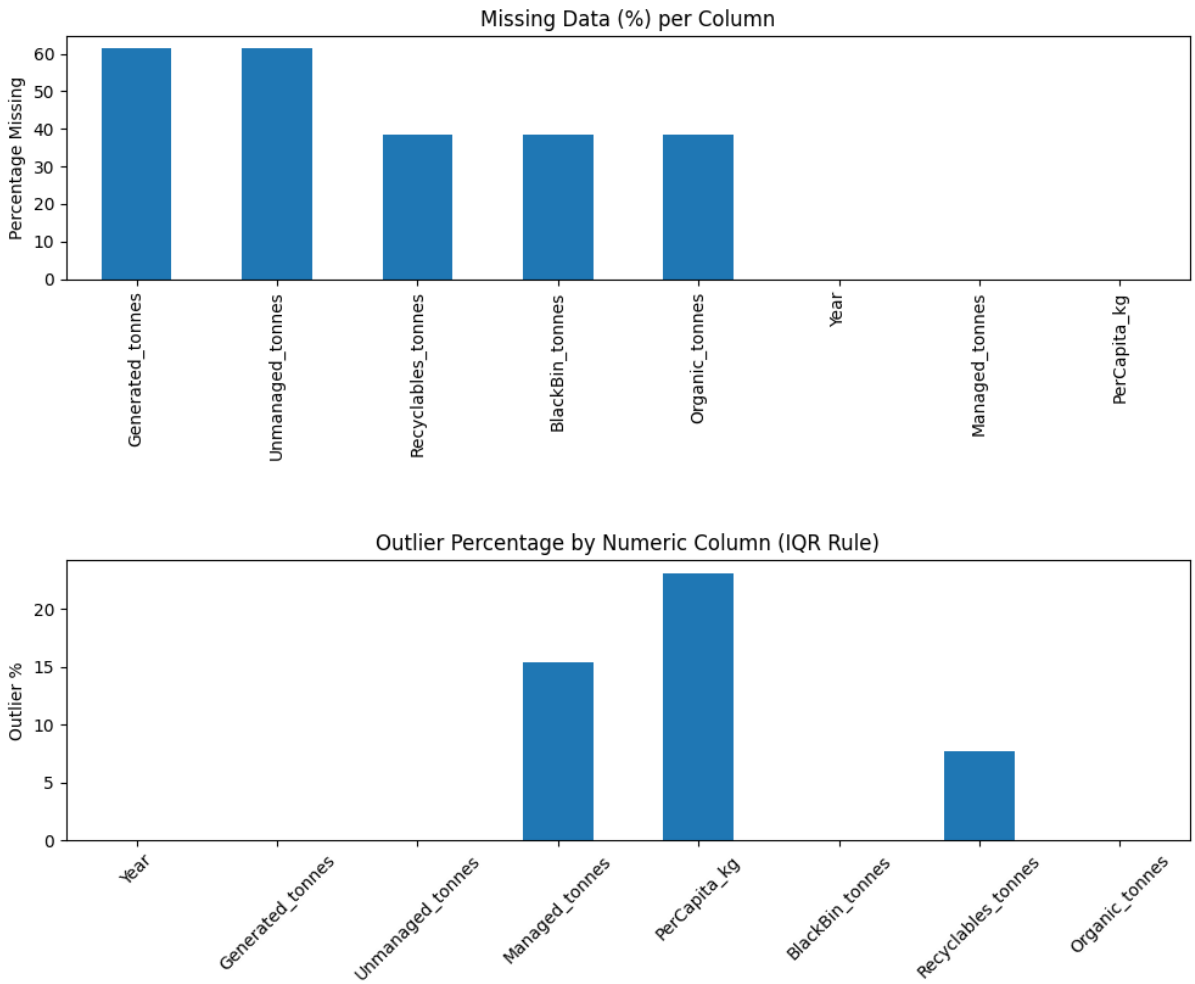



*Figure 4: Missing % and Outlier % for EPA Household Dataset*

## 6.3 EPA Compost and Anaerobic Digestion

This bio-waste data is characterized by an extremely high level of missingness and this is a problem here as it prevents the statistical description. The average Bio waste of acceptance stands at about 185k tonnes. However, 95 percent of missing values can be found in such significant variables as Compost_wet_t and AD_EU_T, therefore, it is not possible to compute the measures of dispersion. The limited data that was available did not depict any outliers (EPA, 2023).

```
data3.describe()
```

| | Bio_waste_accepted_t | Compost_collected_t | Compost_wet_t | Compost_EU_t | AD_collected_t | AD_wet_t | AD_EU_t | Total_collected_t | Total_wet_t | Total_EU_t |
|---|---|---|---|---|---|---|---|---|---|---|
| count | 18.000000 | 5.000000 | 1.0 | 1.0 | 5.000000 | 1.0 | 1.0 | 5.000000 | 1.0 | 1.0 |
| mean | 184777.777778 | 283930.000000 | 306801.0 | 251805.0 | 229568.400000 | 428703.0 | 257767.0 | 513498.400000 | 735504.0 | 509572.0 |
| std | 94421.181295 | 10310.950344 | NaN | NaN | 81050.036202 | NaN | NaN | 88287.571839 | NaN | NaN |
| min | 48000.000000 | 267205.000000 | 306801.0 | 251805.0 | 120397.000000 | 428703.0 | 257767.0 | 401324.000000 | 735504.0 | 509572.0 |
| 25% | 107000.000000 | 280927.000000 | 306801.0 | 251805.0 | 178136.000000 | 428703.0 | 257767.0 | 445341.000000 | 735504.0 | 509572.0 |
| 50% | 169000.000000 | 288695.000000 | 306801.0 | 251805.0 | 237637.000000 | 428703.0 | 257767.0 | 528082.000000 | 735504.0 | 509572.0 |
| 75% | 254250.000000 | 290445.000000 | 306801.0 | 251805.0 | 303176.000000 | 428703.0 | 257767.0 | 595554.000000 | 735504.0 | 509572.0 |
| max | 334000.000000 | 292378.000000 | 306801.0 | 251805.0 | 308496.000000 | 428703.0 | 257767.0 | 597191.000000 | 735504.0 | 509572.0 |

*Figure 5: Summary Statistics for EPA Compost and Anaerobic Digestion Dataset*

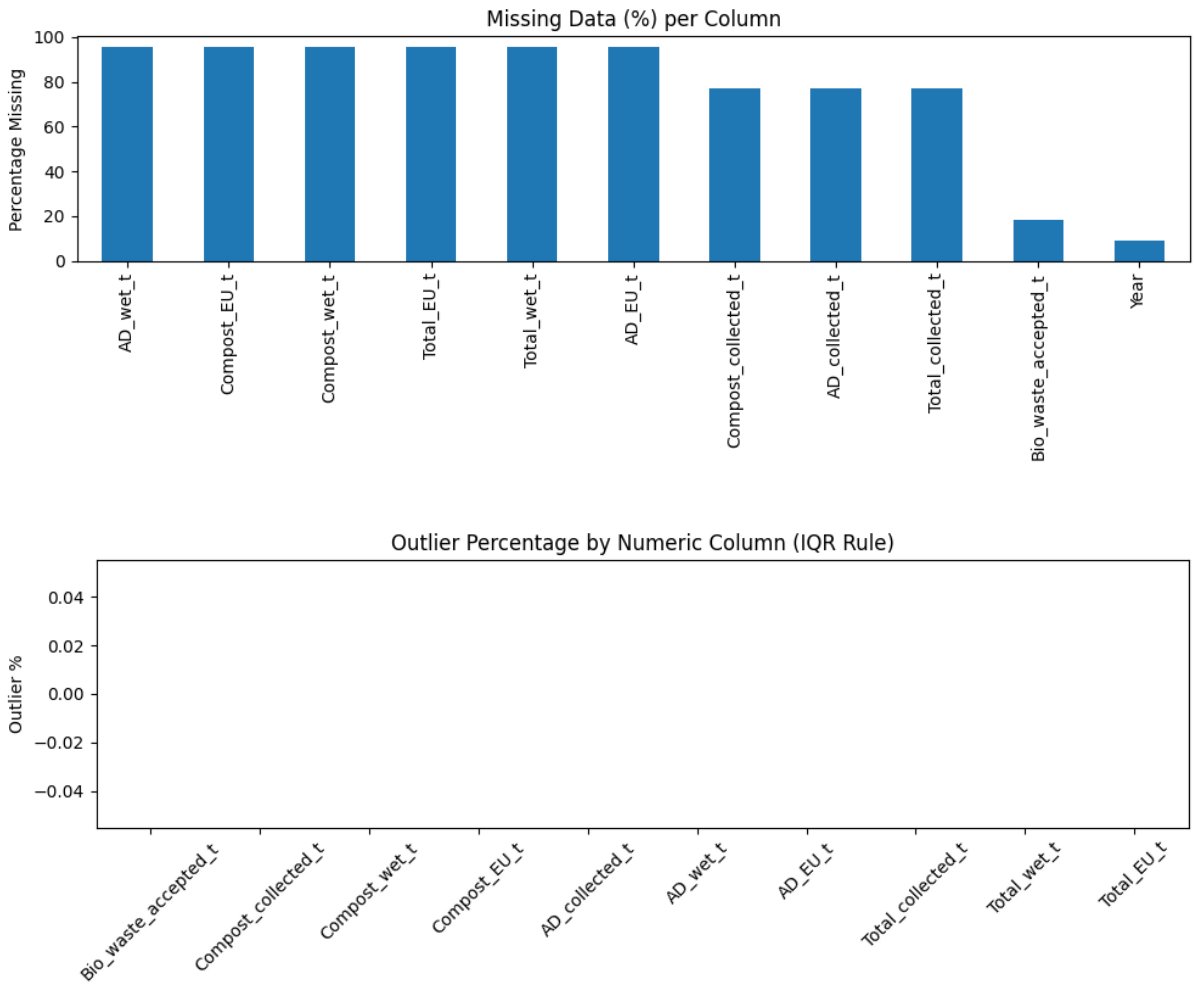



*Figure 6: Missing % and Outlier % for EPA Compost and Anaerobic Digestion Dataset*

## 6.4 Food Waste

This critical dataset (2018-2023) has a mean total food waste of ~877k tonnes. The processing/manufacturing stage is the largest contributor (mean ~327k tonnes), followed by households (mean ~235k tonnes). Distributions are generally balanced (mean ≈ median), and standard deviations show moderate variability. Missing data is minimal overall, except for *Primary_Production* (33% missing). Outliers are limited to one each in the *Primary_Production* and Restaurants categories (EPA, 2023).

| | Year | Total_t | Primary_Production_t | Processing_Manufacturing_t | Retail_Distribution_t | Restaurants_FoodService_t | Households_t |
|---|---|---|---|---|---|---|---|
| **count** | 6.000000 | 6.000000e+00 | 4.000000 | 6.000000 | 6.00000 | 6.000000 | 6.000000 |
| **mean** | 2020.500000 | 8.768702e+05 | 56567.250000 | 327444.833333 | 86751.00000 | 189903.833333 | 235059.000000 |
| **std** | 1.870829 | 1.586260e+05 | 9321.918664 | 135713.031124 | 17940.82745 | 27512.614346 | 16405.813019 |
| **min** | 2018.000000 | 7.492430e+05 | 50032.000000 | 215000.000000 | 60900.00000 | 157045.000000 | 220158.000000 |
| **25%** | 2019.250000 | 7.573250e+05 | 52135.750000 | 222150.000000 | 77201.50000 | 175911.000000 | 220963.250000 |
| **50%** | 2020.500000 | 8.026290e+05 | 52918.500000 | 267610.500000 | 86454.50000 | 183750.000000 | 231000.000000 |
| **75%** | 2021.750000 | 9.990145e+05 | 57350.000000 | 449366.250000 | 97575.75000 | 199725.000000 | 249625.000000 |
| **max** | 2023.000000 | 1.100020e+06 | 70400.000000 | 497500.000000 | 111297.00000 | 236530.000000 | 254745.000000 |

*Figure 7: Summary Statistics for EPA Food Waste Dataset*

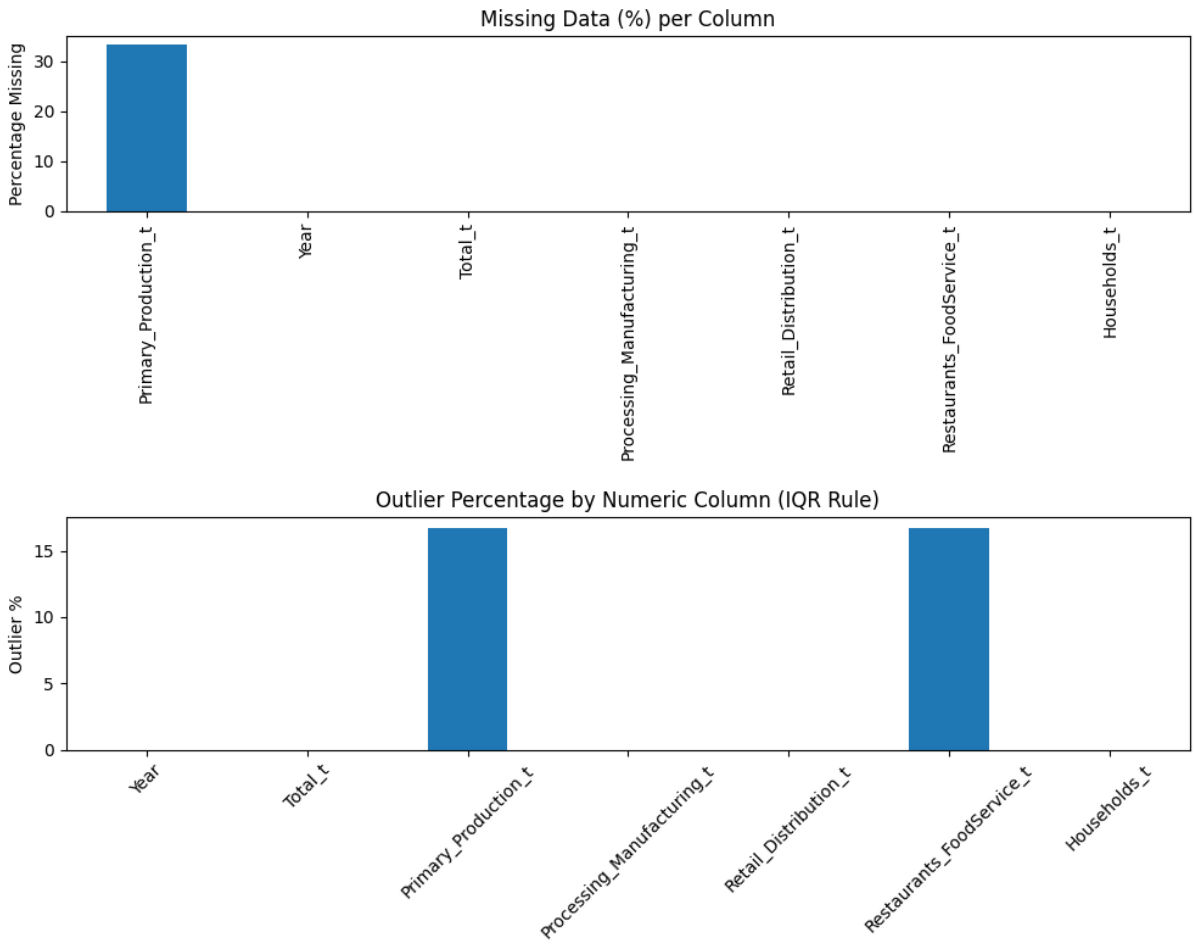



*Figure 8: Missing % and Outlier % for EPA Food Waste Dataset*

## 6.5 EEA GHG Emissions Dataset

After filtering for Ireland, the main numeric variable is emissions. The mean value is approximately 0.61 units (dependent on the filtered subset). A critical issue is that the emissions field itself has 39.3% missing values, a major gap for analysis. Furthermore, ~12.3% of the available emission values are statistical outliers, likely representing extreme events or reporting anomalies (EEA, 2025).

| | Year | emissions | PublicationDate |
|---|---|---|---|
| **count** | 629220.000000 | 3.821440e+05 | 629220.0 |
| **mean** | 2006.495218 | 8.259723e+03 | 20250315.0 |
| **std** | 9.808253 | 1.028086e+05 | 0.0 |
| **min** | 1990.000000 | -4.698874e+05 | 20250315.0 |
| **25%** | 1998.000000 | 1.859800e-01 | 20250315.0 |
| **50%** | 2006.000000 | 1.935670e+01 | 20250315.0 |
| **75%** | 2015.000000 | 6.082974e+02 | 20250315.0 |
| **max** | 2023.000000 | 4.872571e+06 | 20250315.0 |

*Figure 9: Summary Statistics for EEA GHG Emissions Dataset*

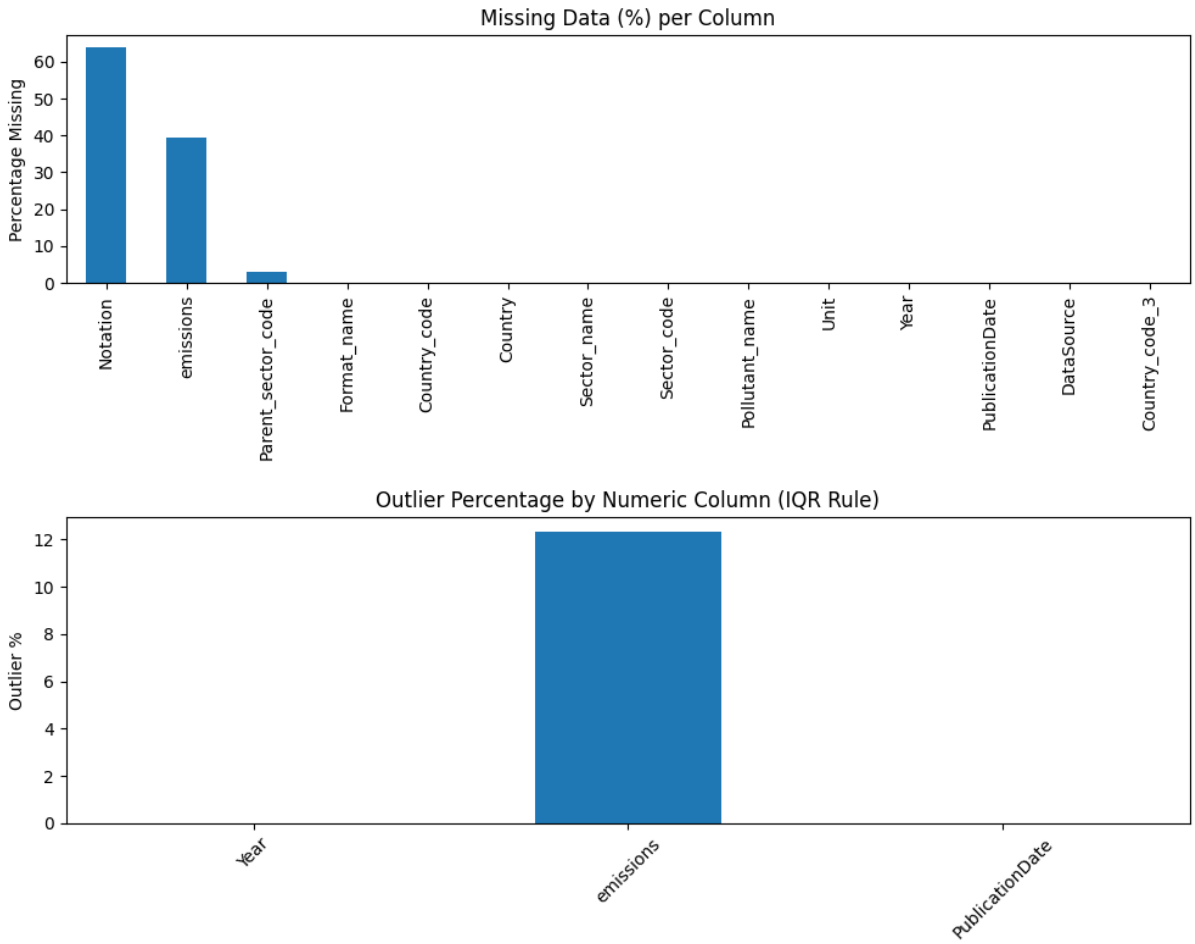


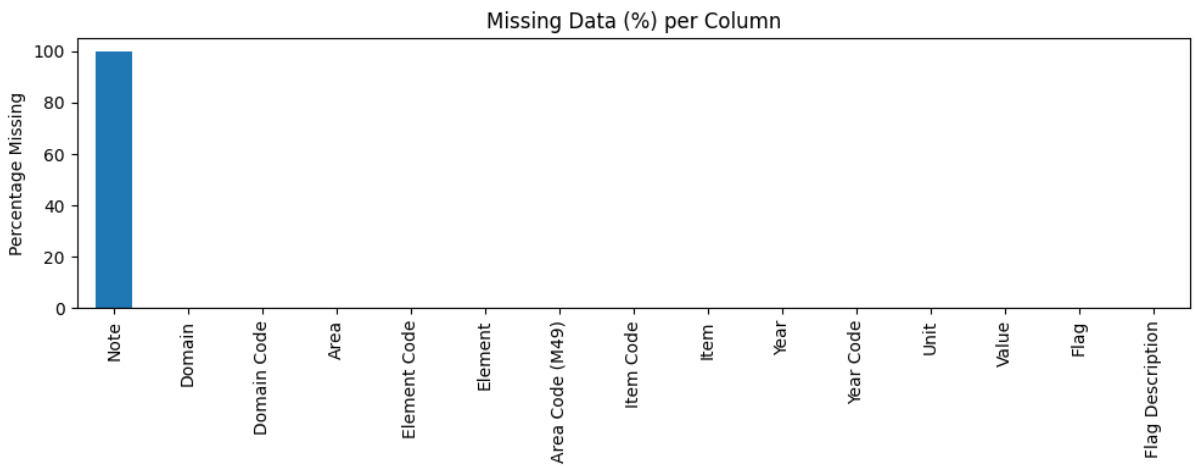

*Figure 10: Missing % and Outlier % for EEA GHG Emissions Dataset*

## 6.6 FAO Annual Population

This dataset (1950-2023) is complete and consistent. The mean population is ~3,640 thousand, with a slightly lower median (~3,521 thousand), reflecting steady growth. The standard deviation of ~715 thousand and the range from ~2,825 to ~5,197 thousand confirm this gradual increase over decades. There are no missing values in the key Value field and no outliers (FAO, 2023).

| | Area Code (M49) | Element Code | Item Code | Year Code | Year | Value | Note |
|---|---|---|---|---|---|---|---|
| **count** | 74.0 | 74.0 | 74.0 | 74.000000 | 74.000000 | 74.000000 | 0.0 |
| **mean** | 372.0 | 511.0 | 3010.0 | 1986.500000 | 1986.500000 | 3639.984608 | NaN |
| **std** | 0.0 | 0.0 | 0.0 | 21.505813 | 21.505813 | 714.860842 | NaN |
| **min** | 372.0 | 511.0 | 3010.0 | 1950.000000 | 1950.000000 | 2824.693000 | NaN |
| **25%** | 372.0 | 511.0 | 3010.0 | 1968.250000 | 1968.250000 | 2940.930500 | NaN |
| **50%** | 372.0 | 511.0 | 3010.0 | 1986.500000 | 1986.500000 | 3520.524500 | NaN |
| **75%** | 372.0 | 511.0 | 3010.0 | 2004.750000 | 2004.750000 | 4132.979000 | NaN |
| **max** | 372.0 | 511.0 | 3010.0 | 2023.000000 | 2023.000000 | 5196.630000 | NaN |

*Figure 11: Summary Statistics for FAO Annual Population*



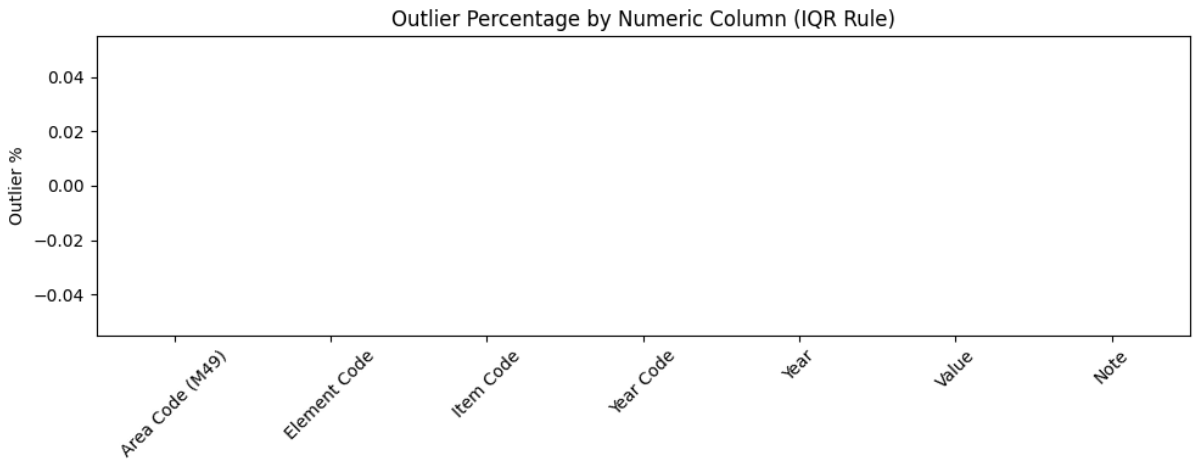



*Figure 12: Missing % and Outlier % for FAO Annual Population*

## 6.7 FAO Food Production Indices

This dataset (1961-2023) shows a highly skewed distribution. The mean index value is ~206, but the median is only ~96, indicating the mean is heavily influenced by extremely high values. This is confirmed by a very high standard deviation (~608) and a vast range from 0.83 to 14,480.68. Approximately 13% of the index values are statistical outliers, which will significantly impact any analysis using averages (FAO, 2023).

| | Area Code (M49) | Element Code | Year Code | Year | Unit | Value |
|---|---|---|---|---|---|---|
| **count** | 4666.0 | 4666.000000 | 4666.000000 | 4666.000000 | 0.0 | 4666.000000 |
| **mean** | 372.0 | 433.000000 | 1992.417917 | 1992.417917 | NaN | 205.943311 |
| **std** | 0.0 | 1.000107 | 17.128723 | 17.128723 | NaN | 608.301323 |
| **min** | 372.0 | 432.000000 | 1961.000000 | 1961.000000 | NaN | 0.830000 |
| **25%** | 372.0 | 432.000000 | 1978.000000 | 1978.000000 | NaN | 70.252500 |
| **50%** | 372.0 | 433.000000 | 1993.000000 | 1993.000000 | NaN | 96.160000 |
| **75%** | 372.0 | 434.000000 | 2007.000000 | 2007.000000 | NaN | 131.292500 |
| **max** | 372.0 | 434.000000 | 2023.000000 | 2023.000000 | NaN | 14480.680000 |

*Figure 13: Summary Statistics for FAO Food Production Indices Dataset*

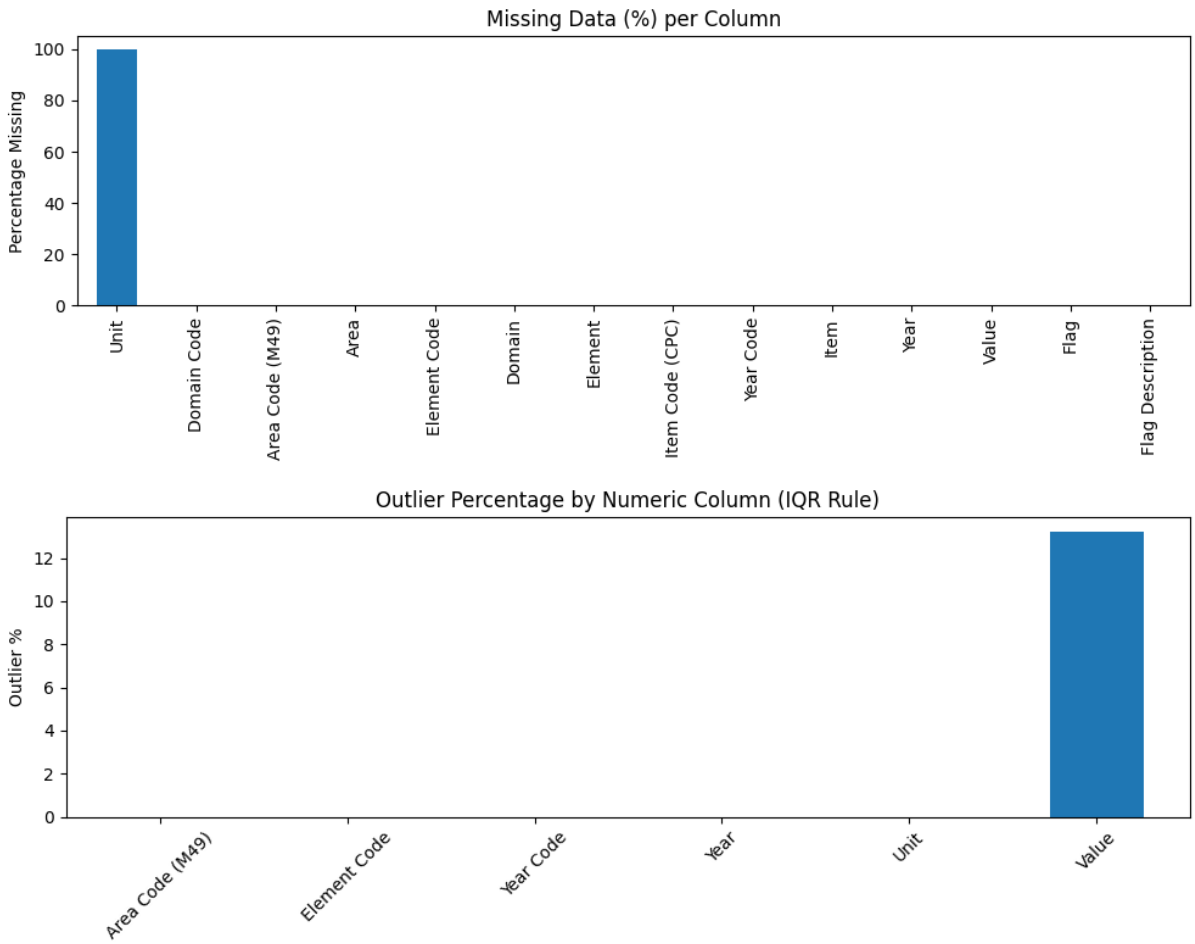



*Figure 14: Missing % and Outlier % for FAO Food Production Indices Dataset*

## 6.8 Summary of Key Quantitative Characteristics

| Dataset | Key Variable(s) | Central Tendency | Dispersion | Missingness | Outliers |
|---|---|---|---|---|---|
| **EPA Municipal** | Generated Tonnes | Mean ~2.9M t | Std ~650k t | **High (~61%)** | Minimal (in recycling) |
| **EPA Household** | Generated Tonnes | Mean ~1.55M t | Std ~184k t | **High (~62%)** | Moderate (in per-capita) |
| **Compost & AD** | Bio-waste Accepted | Mean ~185k t | N/A (High Missingness) | **Very High (~77-95%)** | None |

| | | | | | |
|---|---|---|---|---|---|
| **Food Waste** | Total Food Waste | Mean ~877k t | Std ~150k t | **Low (excl. Primary Prod.)** | Minimal |
| **EEA GHG** | Emissions | Mean ~0.61 units | Varies | **Very High (39.3%)** | **High (~12.3%)** |
| **FAO Population** | Value | Mean ~3,640k | Std ~715k | **None** | None |
| **FAO Food Indices** | Value | Mean: 206, Median: 96 | Std ~608 | **None** | **High (~13%)** |

# 7. Data Processing for Dashboard Creation

The data was processed using a two-stage pipeline: initial cleaning and transformation in Python, followed by advanced metric creation and visualisation in Power BI using DAX (Data Analysis Expressions). This combined approach ensured robust data preparation and dynamic, user-friendly analytics.

## 7.1 Foundational Processing: Structuring and Cleaning

All raw datasets underwent standardised preprocessing in Python to create a clean, consistent foundation.

- Standardisation: Column names were renamed for clarity and consistency (e.g., "Municipal waste managed (tonnes)" became Managed_tonnes).
- Data Quality Analysis: Missingness percentages were calculated, and simple outlier detection was performed using the Interquartile Range (IQR) method to understand data integrity.
- Handling Missing Values: For time-series waste data (EPA Municipal, Household, Compost), missing numeric values were filled using linear interpolation to preserve temporal trends. In the EEA GHG dataset, rows with missing emissions values were removed entirely, as imputation was deemed unsuitable for this metric (EPA, 2023).
- Filtering and Integration: The EEA dataset was filtered for relevant Irish sectors and subsectors. Crucially, the Food Waste and FAO Population datasets were joined to calculate the core food_waste_per_capita_kg metric ((Total_Food_Waste_tonnes / Population) * 1000). Sectors within the Food Waste dataset were also grouped into Household and Commercial categories for clearer analysis.

## 7.2 Feature Engineering for Enhanced Analysis

Processed data was loaded into Power BI, where DAX was used to create calculated columns and measures for dynamic analysis.

### 7.2.1 Dashboard 1: SDG 12.3 Progress & National Waste System

SDG 12.3 Progress Metrics: Key measures track the primary goal of halving per capita food waste:

- Baseline_2015, Current_2023, and Target_2030 ([Baseline_2015] * 0.5) provide the core summary for KPI cards.

```
Baseline_2015 =
    CALCULATE(
        MAX(food_waste_per_capita[food_waste_per_capita_kg]),
        food_waste_per_capita[Year] = 2015
    )
```

```
Current_2023 =
    CALCULATE(
        MAX(food_waste_per_capita[food_waste_per_capita_kg]),
        food_waste_per_capita[Year] = 2023
    )
```

```
Target_2030 = [Baseline_2015] * 0.5
```

- Historical_Reduction_Rate and Required_Reduction_Rate quantify past performance and the accelerated pace needed to meet the 2030 target.

```
Historical_Reduction_Rate =
    DIVIDE([Baseline_2015] - [Current_2023], 8, 0)
```

```
Required_Reduction_Rate =
    DIVIDE([Current_2023] - [Target_2030], 7, 0)
```

- A calculated table (SDG_Trajectory) generates the ideal linear path from 2015 to 2030, enabling clear visual gap analysis in the central line chart.

```
SDG_Trajectory =
VAR BaselineYear = 2015
VAR TargetYear = 2030
VAR ReductionFraction = 0.5

VAR BaselineValue =
    CALCULATE(
        MAX(food_waste_per_capita[food_waste_per_capita_kg]),
        food_waste_per_capita[Year] = BaselineYear
    )

VAR TargetValue = BaselineValue * (1 - ReductionFraction)

RETURN
ADDCOLUMNS(
    GENERATESERIES(BaselineYear, TargetYear, 1),
    "Trajectory_Value",
        BaselineValue -
        (([Value] - BaselineYear) / (TargetYear - BaselineYear)) *
        (BaselineValue - TargetValue)
)
```

Waste System Efficiency: For the Municipal dataset, DAX measures calculate the annual percentage share of total managed waste treated by each method:

- Landfill_Percent, Recycling_Percent, Energy_Recovery_Percent, and Organic_Diversion_Percent. (The same formula repeated for other variables too).

```
Energy_Recovery_Percent =
    DIVIDE(
        SUM(Municipal_data[Energy_Recovery_tonnes]),
        SUM(Municipal_data[Managed_tonnes])
    ) * 100
```

- These feed the Stacked Area Chart showing the evolution of waste management. Corresponding Baseline_ and Current_ measures (e.g., Baseline_Landfill_Rate, Current_Landfill_Rate) populate KPI cards to highlight changes since 2015. (The same formula repeated for other variables too).

```
Baseline_Landfill_Rate =
    CALCULATE(
        [Landfill_Percent],
        Municipal_data[Year] = 2015
    )
```

```
Current_Landfill_Rate =
    CALCULATE(
        [Landfill_Percent],
        Municipal_data[Year] = 2023
    )
```

### 7.2.2 Dashboard 2: Carbon & Social Impact

Climate Impact (SDG 13): Measures were created to link food waste to greenhouse gas emissions.

- Current_Emissions_tonnes and Emissions_Avoided (vs. 2015) provide high-level carbon KPIs.

```
Current_Emissions_tonnes =
    CALCULATE(
        SUM(EEA_Ireland_emissions_data[CO2_equivalent_tonnes]),
        EEA_Ireland_emissions_data[Year] = 2023
    )
```

```
Emissions_Avoided =
    VAR Baseline = CALCULATE(SUM(EEA_Ireland_emissions_data[CO2_equivalent_tonnes]), EEA_Ireland_emissions_data[Year] = 2015)
    VAR Current1 = [Current_Emissions_tonnes]
    RETURN Baseline - Current1
```

- Emission_Intensity (Total Emissions / Total Food Waste) calculates the carbon footprint per tonne of waste, tracked over time in a line chart.

```
Emission_Intensity =
    DIVIDE(
        SUM(EEA_Ireland_emissions_data[CO2_equivalent_tonnes]),
        SUM(food_waste_per_capita[Total_t])
    )
```

- Sector-specific measures (e.g., Anaerobic_at_Biogas_Emissions, Food_Processing_Emissions) enable the breakdown in the Pie Chart, identifying key emission sources.

```
Anaerobic_at_Biogas_Emissions =
    CALCULATE(
        SUM(EEA_Ireland_emissions_data[CO2_equivalent_tonnes]),
        EEA_Ireland_emissions_data[Sector_name] = "5.B.2 - Anaerobic Digestion at Biogas Facilities"
    )
```

```
Composting_Emissions =
    CALCULATE(
        SUM(EEA_Ireland_emissions_data[CO2_equivalent_tonnes]),
        EEA_Ireland_emissions_data[Sector_name] = "5.B.1 - Waste Composting"
    )
```

```
Food_Processing_Emissions =
    CALCULATE(
        SUM(EEA_Ireland_emissions_data[CO2_equivalent_tonnes]),
        EEA_Ireland_emissions_data[Sector_name] = "1.A.2.e - Food Processing, Beverages and Tobacco"
    )
```

```
Manufacturing_Emissions =
    CALCULATE(
        SUM(EEA_Ireland_emissions_data[CO2_equivalent_tonnes]),
        EEA_Ireland_emissions_data[Sector_name] = "1.A.2 - Manufacturing Industries and Construction"
    )
```

```
Open_Burning_Emissions =
    CALCULATE(
        SUM(EEA_Ireland_emissions_data[CO2_equivalent_tonnes]),
        EEA_Ireland_emissions_data[Sector_name] = "5.C.2 - Open Burning of Waste"
    )
```

Social Impact (SDG 2): To contextualise waste in human terms, a "People Fed" metric was developed.

- Meals_Equivalent is a calculated measure based on the hypothesis that 70% of total food waste is edible. It converts the total edible tonnage into an equivalent number of meals (assuming 0.5kg per meal), creating a powerful, relatable KPI card that frames waste as a missed opportunity to address food insecurity.

```
Meals_Equivalent =
    CALCULATE(((AVERAGE(food_waste_per_capita[Total_t]) * 0.70) * 1000) / 0.5, food_waste_per_capita[Year]=2023)
```

## 7.3 Dataset Simplification and Focus

Intentional simplification was used in order to streamline the data model to achieve the best performance in the dashboard and clarity to the user. Unnecessary metadata columns which were not central to the analytical story were automatically dropped. In general, the dataset showed the results of EPA Compost and Anaerobic Digestion were simplified to only two key metrics: Bio waste accepted t and Compost collected t. Other supporting or assisting columns were omitted so as to make the analysis of the intake and processing volumes of organic waste to be the focal point.

- FAO Datasets: The dataset was cleaned of all the unnecessary metadata, country codes and flags. The basic Year and Value columns were kept only, which resulted in a simple and lightweight table with an excellent design to compute per capita.
- FAO Food Production Indices: This data was eventually excluded in the final analysis. Although it was originally regarded to offer some context on agriculture, the index values did not have any apparent and actionable unit of measurement (e.g., tonnes, kcal) that could be directly compared or correlated with the weight of food waste. The indices failed to provide some clarity and interpretable context to the SDG 12.3 narrative and were removed to ensure analytical accuracy to prevent possible misinterpretation without a clear conversion method. •

- EEA Emissions Dataset: Having filtered by country (Ireland) and sector (relevant sectors), many of the detailed classification columns (e.g. notation, parent codes) were dropped. The dataset was reduced to essential variables: Year, Sector_name, Pollutant_name and CO2 equivalent tonnes.

This simplification process made the final data model lean, consisting of only the metrics needed to populate the KPIs and visualisations of the dashboard, which enhanced the load times, consumer cognitive load of report developers, and the risk of confusion caused by useless data fields.

# 8. Dashboard Visualisation and Design Rationale

The dashboard was designed using core visualisation principles to ensure clarity, accuracy, and ease of understanding for a diverse client audience at GreenPlate Ireland. The layout is structured to tell a logical story: from tracking the core SDG 12.3 target, to analysing the supporting national waste system, and finally, to quantifying the cross-cutting climate and social impacts (United Nations, 2025).

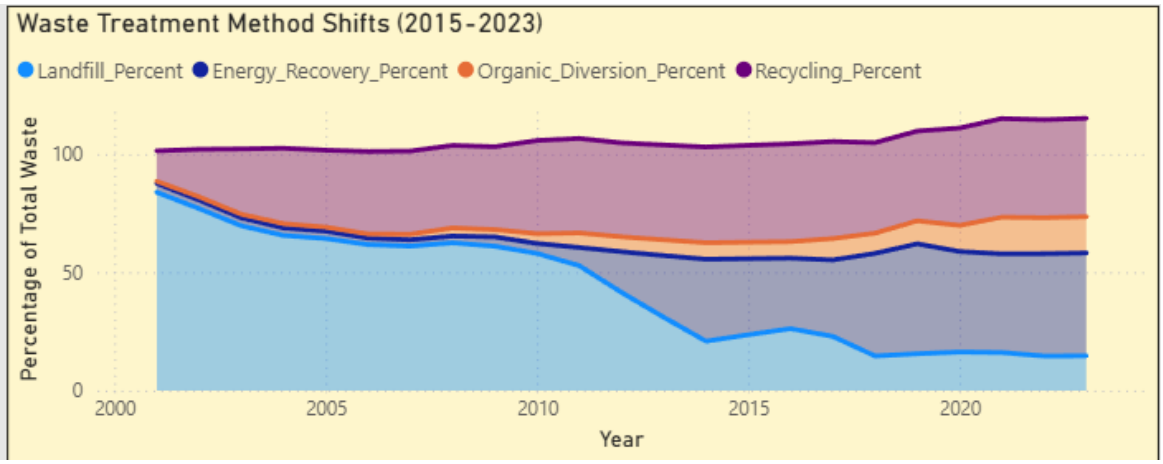
## 8.1 Rationale for Graph Choices

The graph selections priorities the effective communication of different data types and relationships, aligning specific chart types with the analytical question each visual aim to answer.

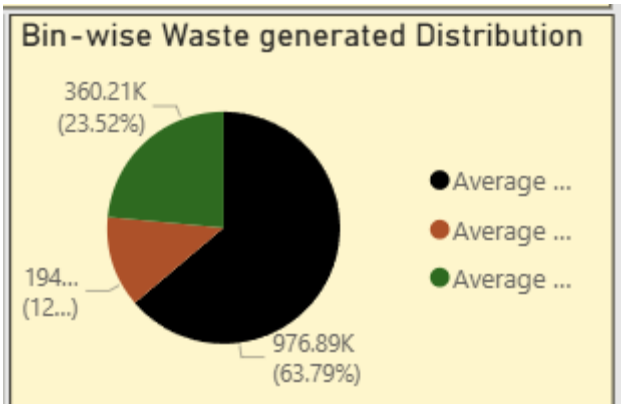
### 8.1.1 Dashboard 1: SDG Progress & National Waste System

SDG Trajectory Chart (Line Graph): This is the core illustration of monitoring the progress against SDG 12.3. A line graph is best when the trends are to be represented. It shows three series on the X-axis (Year) and Y-axis (kg per capita) the Actual historical trend, the ideal linear SDG Target Path to 2030 and the Current Trajectory projection. This can provide a gap analysis briefly. Bar chart was also thought about and this was rejected because it is a bad measure to depict the continuous target and future projection.

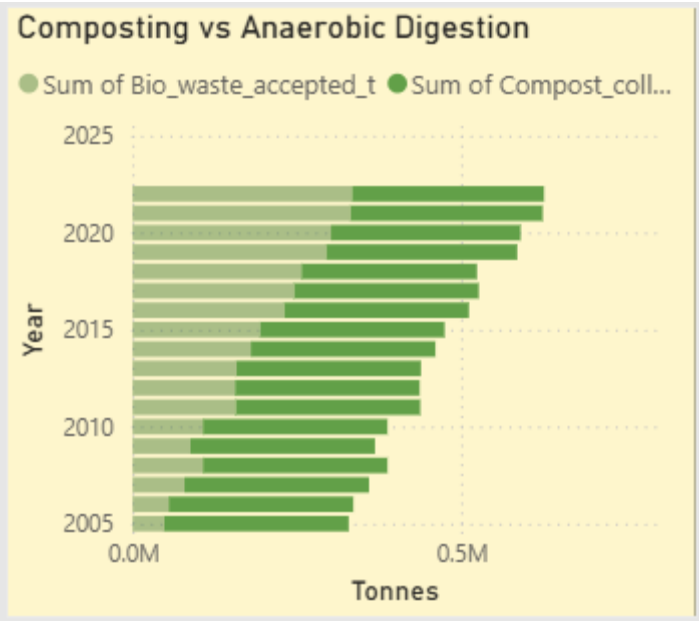
- Shifts in *Treatment Method (Stacked Area Chart)*: The graph is used to visualize the changing composition of the municipal waste treatment. The stack area chart was selected to clearly demonstrate the part and whole correlation with respect to time passing (X-axis: year, Y-axis: percentage). It brings out the change of shares in land fill, energy recovery, recycling and organic diversion and the transformation between disposal and recovery is well defined. A grouped bar chart was abandoned because it hides the amount of waste that is being managed, and it becomes hard to compare the trends.




- *Household Bin Waste (Pie Chart)*: In the case of the latest year, a pie chart will be able to demonstrate the proportion dissolution of household waste to General (Black Bin), Recycling, and Organic streams. It is a straightforward, simple format that gives a snapshot of composition at the moment and gets the answer to the question What is in our bins?.



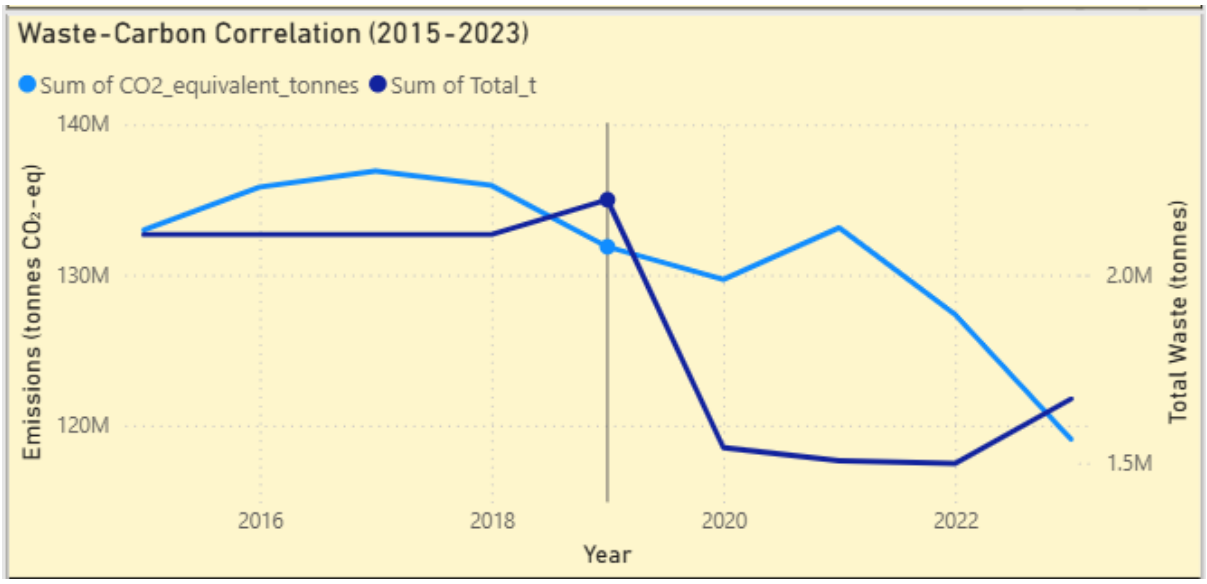

- *Composting vs. Anaerobic Digestion* (Stacked Bar Chart): As a vertical stacked bar chart (X-axis: Year, Y-axis: Tonnes), the annual yearly amount of biowaste received and its division between composting and anaerobic digestion (AD) are evident.




This conveys well the development and the proportionate uptake of these two important treatment technologies as time goes by.

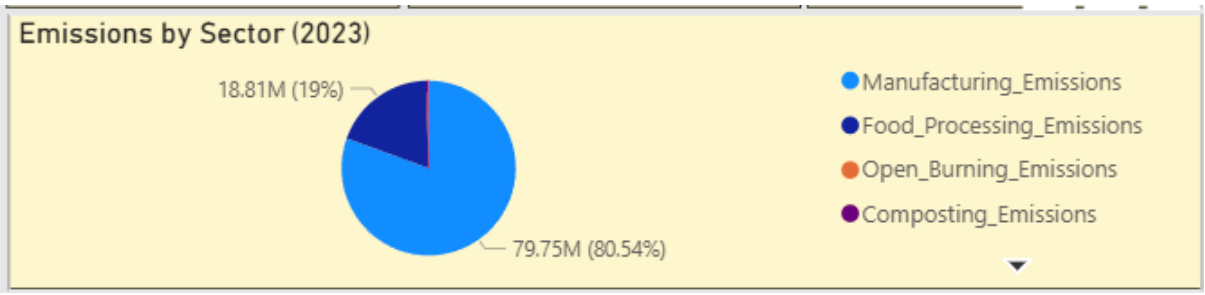
### 8.1.2 Dashboard 2: Carbon & Social Impact

*Waste-Carbon Correlation (Dual-Axis Line Chart)*: This is a chart that shows two related time series on a common X-axis (Year): total CO 2 equal emissions and total food waste (tonnes). The thrust of the question of whether the trends in emissions are in line with food waste can meaningfully be answered by using dual Y-axis to directly correlate their trend. A scatter chart was also an option, but a line chart is more effective in focusing on the trend with time with each metric.

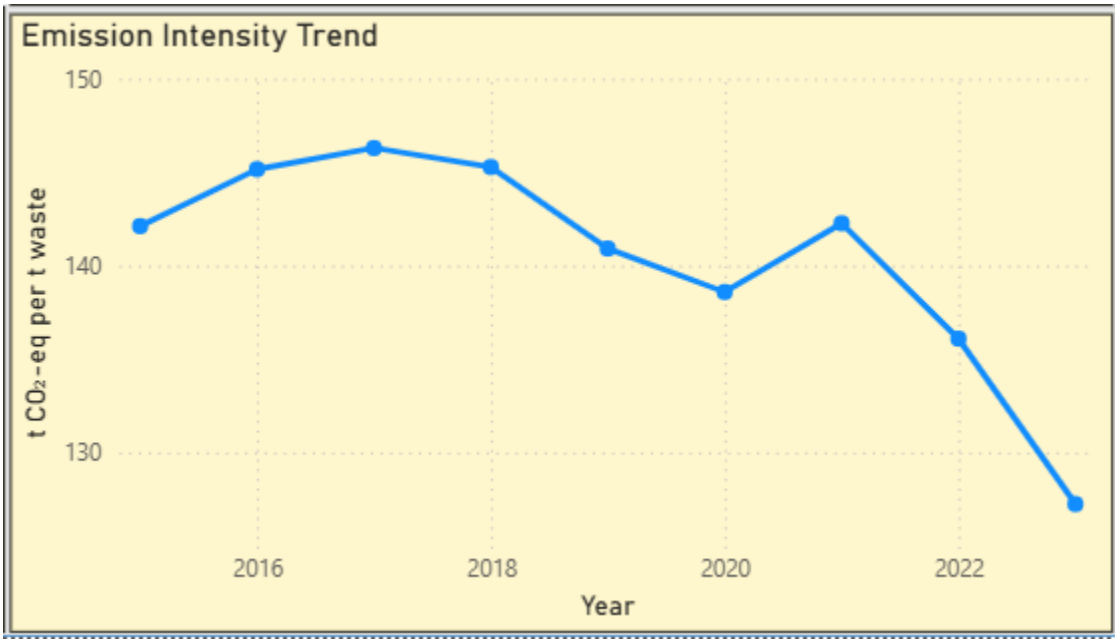



Emissions by Sector (Pie Chart): This chart is a static annual breakdown of total emissions showing percentages contributed by each of the main contributing sectors, such as landfill, food processing and waste composting / AD. A pie chart is suitable in this case to be able to

judge a part-whole comparison in a basic and intuitive manner and be able to distinguish the largest contributing segments at first sight.




*Emission Intensity Trend (Line Chart):* This is a single-axis line chart, which follows the calculated value of emission per tonne of food waste. It provides the key question of the day, the critical efficiency question: Is the carbon footprint of every tonne of waste increasing or decreasing? The obvious way of presenting this continuous trend is through a line chart.




## 8.2 Rationale for Overall Dashboard Layout and Design

The overall design adheres to principles of visual hierarchy, consistency, and guided narrative to ensure clarity and actionable insight.

## 8.2.1 Dashboard 1: Logical, Two-Panel Narrative

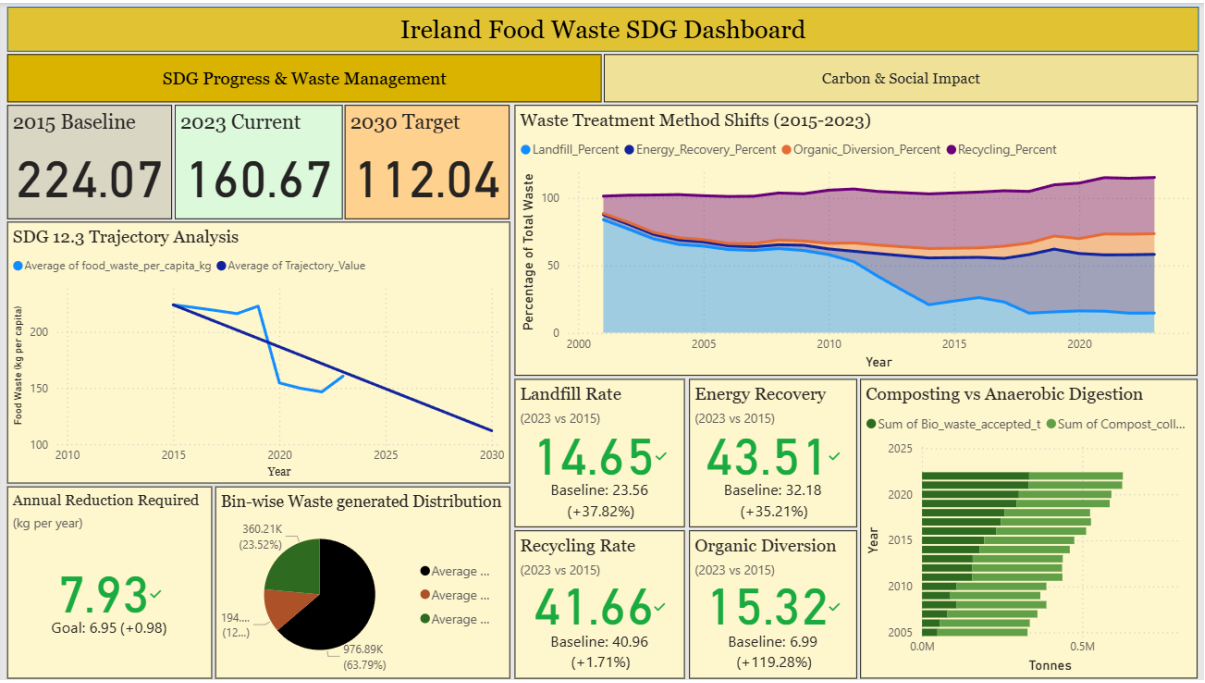



*Figure 15: Dashboard 1 - SDG Progress & Waste Management*

The layout is intentionally split into two clear panels, creating a logical "diagnostic" flow.

- Left Panel (The Goal & Source): This side is dedicated to SDG 12.3 Progress. It begins with the three Display Cards (Baseline, Current, Target) for an instant high-level summary. Directly below, the SDG Trajectory Chart provides the detailed trend analysis, followed by the supporting Annual Reduction KPI and the Household Bin pie chart identifying the source of one major waste stream. This sequence tells the story: *Here is our target, here is our progress, and here is what households contribute.*
- Right Panel (The System Response): This side visualizes the National Waste Management System. The prominent Stacked Area Chart shows the macro-level treatment shifts, with supporting KPI Cards (Landfill Rate, etc.) highlighting key changes since the baseline. The Composting vs. AD chart provides a focused drill-down on organic treatment. This panel answers: *How is the system managing the waste, and how is it improving?*
- Strategic Use of Cards: KPI cards are used consistently to spotlight singular, critical metrics (e.g., a rate, a required reduction). Their placement directly below related complex charts (like the stacked area chart) allows viewers to absorb the detailed trend first, then focus on a specific, derived takeaway.

### 8.2.2 Dashboard 2: Cause-and-Effect Impact Analysis

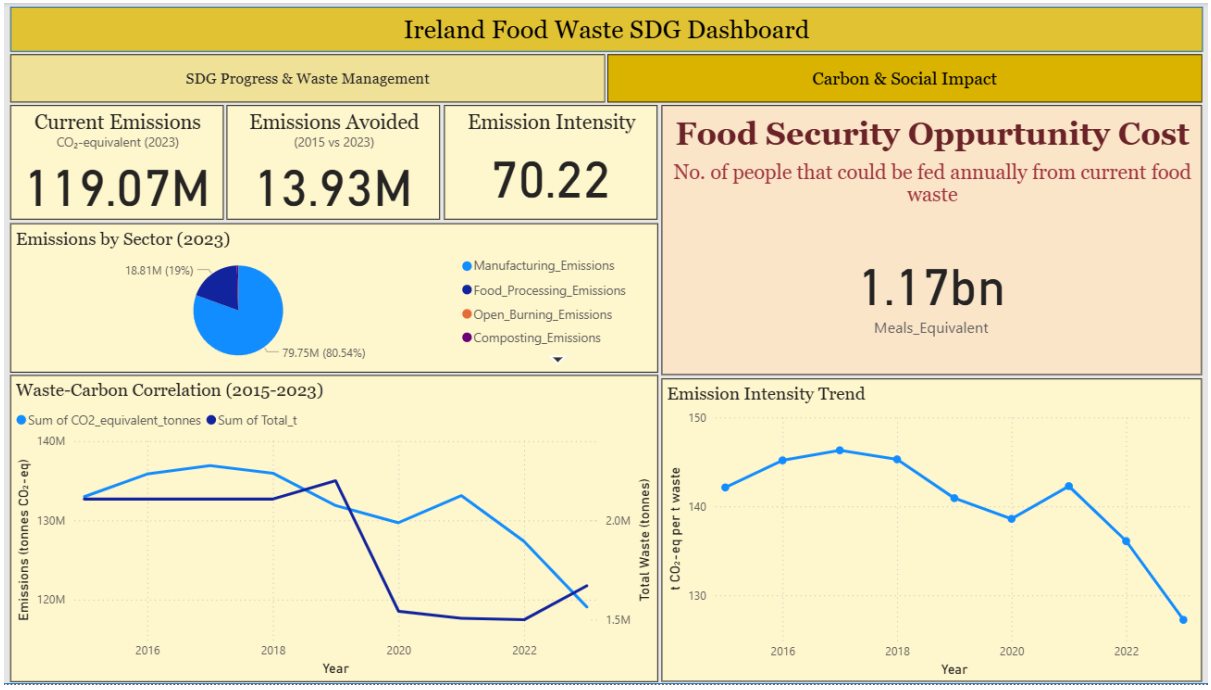



*Figure 16: Dashboard 2 - Carbon & Social Impact*

This dashboard is structured to first quantify the problem, then understand it.

- Top Row (Climate Impact Quantified): The three Carbon Display Cards (Current Emissions, Emissions Avoided, Emission Intensity) provide the immediate top-line climate impact. Directly below, the Waste-Carbon Correlation line chart visually proves the relationship, and the Emissions by Sector pie chart identifies the culprits. This creates a compelling cause-and-effect chain.
- Bottom Row (Social Impact Context): The large Meals Equivalent Card translates abstract tonnes into a powerful, human-centric metric, directly linking to SDG 2 (Zero Hunger). Alongside it, the Emission Intensity Trend line chart shows whether the system's carbon efficiency is improving, providing a forward-looking indicator.
- Consistent and Meaningful Colour Scheme: A consistent palette is applied across both dashboards. In the waste treatment area chart, landfill is in an intuitive "negative" grey/red, while recovery methods are in "positive" greens and blues. Colours for the SDG trajectory lines (Actual, Target, Projection) are maintained in related KPI cards. This consistency reduces cognitive load and aids pattern recognition.

# 8. Recommendations for GreenPlate Ireland

The dashboard analysis shows outstanding evidence that Ireland is not on track towards achieving SDG 12.3. To speed up the process, we would suggest a narrow, data-driven approach to concentrate on the most significant leverage points.

- Introduce a Commercial Sector Intervention: It is evident that the most prevalent stream of data is commercial food waste (processing, retail, hospitality). The use of a scattergun approach will not be effective. To this end, we advise formulation and lobbying to a specific government grant scheme to SMEs in the manufacturing and retailing industries to invest in accurate waste monitoring systems and technologies used in waste prevention with the amount of funding depending on the reduction that is shown year by year. Adopt a Carrot and

- Stick Policy on Organic Waste: Although the size of composting volume did not decrease, the dashboard showed that digestion (AD) arose anaerobically despite being better at producing renewable energy. We recommend supporting a policy package whereby there will be incentives to divert commercial organic waste to AD (e.g. by tax credits) and the levy on biodegradable waste in landfills should be raised. This forms a strong economic indicator, promoting SDG 12.3 and SDG 13.

- Implement a Shareable, Metric-Based Social Movement: "Meals Equivalent" is an incredibly effective metric translating abstract tonnage into a social cost. GreenPlate would offer to introduce the campaign of Food, Not Waste and base its message on this measure (e.g. message would be Last year, we wasted Y and this would have served X people). This must be combined with a simplified, standardized national bin labeling system to minimize household confusion which will go right to the root of the lack of organic diversion rates in the household bin analysis.

# 9. Conclusion

This dashboard offers an objective, factual overview of the Ireland development in SDG 12.3. The analysis proves that there is indeed a performance gap, but there exist accurate and high-impact areas of intervention. GreenPlate Ireland can turn these lessons into action plans by dedicating strategic efforts to the business sector, streamlining the policy of organic waste, and initiating a relatable public campaign. This will help the organization spearhead a focused initiative to get the national food system back on track and transform data into decisive action to create a more sustainable and equitable future.

# Appendix A: References

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# Appendix B: AI Usage Disclosure

The technology of AI was used ethically and professionally in this project to assist us during our design thinking and ideation. The assistive tool that was used was NotebookLM, but not to generate final content, but to explore the design possibilities and speed up the early stage thinking process, maintaining the complete academic integrity and transparency of the process.

**How AI Was Used**

- NotebookLM was used to upload and review the project rubric and related documentation. This helped our team clearly understand expectations, evaluation criteria, and scope, ensuring alignment before starting the design and analysis work. [Link]
- We used Claude.ai to explore high-level design ideas for a dashboard based on an existing food waste dataset. The tool helped us visualize possible layouts and components, which we then critically assessed and refined ourselves. [Link]

- AI was occasionally used during early brainstorming to explore alternative perspectives and prompt creative thinking. All ideas generated were reviewed, filtered, and developed further through team discussion and independent reasoning.
- ChatGPT was used briefly to understand and compare relevant UN Sustainable Development Goals (SDGs) to help us decide which SDG best aligned with current trend. The final selection and justification were made by the team. [LINK]

**Example Prompt:**

"*We have a set of datasets for a food waste report. Based on this data, suggest a mock dashboard view that highlights key metrics and insights.*"

The output served only as an initial reference for layout ideas and did not dictate final design choices.

In general, the artificial intelligence tools were used as a support tool, as opposed to a content-creating tool, and were designed to facilitate understanding, thinking, and productivity. The original team generated all the critical thinking, analytical rigor, design decisions and final outputs, which entailed transparency and adherence to the set standards of academic integrity.

# Tools Used

Google Colaboratory : For Data Preparation and Cleaning : [Link]